Growing Neural Gas for Adaptive Routing Across Evolving Heterogeneous Language Model Pools

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Introduction

As models increasingly rise in architectural structure, the development of LLMs is expanding from monolithic systems to interconnected subsystems of heterogeneous model ecosystems. This stems from the idea that no generalist model exists such that it outperforms specialists in their area of expertise. In 1997, Wolperd et Macready published the No Free Lunch theorem along similar lines, providing a mathematical proof that there is no one model that works best for every situation.

The naïve resolution to this is to query every model followed by comparison of outputs to elect a response. This, however, has high token costs, and scales poorly as models grow in size. The has resulted in LLM routing becoming a distinct subfield.

Over 2024-2026, there has been great growth in exploratory literature around model selection, including RouteLLM (focusing on preferential choice of data), UniRoute (cluster based validation based), and EmbedLLM (converting LLMs into vectorized representations for comparative choicing).

Our method focuses on an unexplored approach to this problem: the application of Growing Neural Gas (GNG), a well-established incremental, self-organizing topology-learning algorithm originating in the unsupervised learning literature, to the construction of a routing structure that grows organically and incrementally from empirical performance data, using explicit geometric reconstruction, providing local interpretability via bucketing and a discrete task expertise representation space.

Problem Statement

Static clustering-based methods of reconstruction often need retraining when models drift, and necessitate retraining. Moreover, some methods fail to define exact task space representations, focusing on ellipsoid-style representations. On the other hand, neural classifiers attain good predictive accuracy, but suffer from black-box paradigms, leading to low debuggability, trust, and auditability.

Moreover, in most literature, fixed model pools are considered, while in reality, it is ideal to change LLMs frequently, occasioned by new releases. With GNG, the added benefit is that changing areas can be isolated by geometric bounds, and only data in that particular dimensional space needs to be changed (leading to less frequent needs for complete retraining in theory).

Our project explores if incremental, self-growing topology-learning algorithm, namely Growing Neural Gas, can be deployed to construct an LLM router that sustains predictive accuracy, adapts efficiently to a continually evolving pool of candidate models, and yields routing decisions that are explainable as an intrinsic consequence of their construction. We aim to apply this to models that were independently grown and trained.

Aim & Objective

Aim: to design, implement, and evaluate a GNG-based router selecting the optimal LLM per prompt from an evolving, heterogeneous pool, emphasizing cost reduction, churn-adaptivity, and explainability.

- Implement a label-aware GNG variant grown over prompt embeddings.
- Incorporate edge-aging/decay so stale regions lose influence without full retraining.
- Compile the topology into a hash-based lookup (LSH) for near constant-time inference.
- Test whether the curse of dimensionality degrades geometry-routing fidelity, and evaluate projection-based mitigations (spatial routers so far suggest the curse of dimensionality does not apply strongly to vector embeddings owing to similarity in clustered topic vocabulary)
- Evaluate against 2 other routing methodologies (one classification, one neural based)

Literature Review

This project situates between three established, yet independently developed, research trajectories, and derives its motivating rationale from a specific gap across the intersection of all three.

MoE

The conceptual foundation for routing inputs to specialized sub-models on the basis of their position within a learned embedding space is traceable to the Mixture of Experts framework, wherein a gating network computes soft, Gaussian-parameterized responsibilities across a constellation of local expert models (Jacobs, Jordan, Nowlan & Hinton, 1991).

Subsequent scholarship has interrogated the replacement of learned gating networks with fixed, deterministic hash functions so as to attenuate routing overhead (Roller, Sukhbaatar, Szlam & Weston, 2021).

Our project's lineage within this class of problems comes from projects like RouteLLM, where an external router is trained outside with heterogeneously trained models versus classic MoE which requires training together. This provides routing at a systems level, relative to MoE's original implementation.

Cluster Based Routing Across Heterogeneous, Independently-Trained LLMs

A distinct and more recent research trajectory addresses routing not among the internal sub-experts of a single model, but among discrete, independently trained LLMs, for instance, the selection among GPT, Claude, and open-weight models for a given query. This trajectory encompasses cluster-then-assign methodologies, which aggregate queries into groups and subsequently assign the empirically best-performing model per group, alongside performance-prediction methodologies, which directly estimate, for each candidate model, its anticipated efficacy on a given query and select the arg-max. Standardized benchmarking suites, such as RouterBench (Hu et al., 2024), have been developed expressly to facilitate consistent evaluation of such routers across a multiplicity of models and datasets.

Explainable Routing

Another area of research within LLM routing focuses on interpretability as a design objective: systems that represent each LLM's competence and each query's difficulty within an explicit, inspectable latent space, and systems that condition routing decisions upon human-legible extracted concepts, such as task type, domain, and reasoning complexity, rather than upon opaque embeddings, are oriented toward rendering individual routing decisions auditable, rather than reliant upon implicit trust in an uninterpretable score. Our project focuses on partial explainability via bucketing topological area definitions.

Incremental Topology Learning

Growing Neural Gas (Fritzke, 1995) constitutes an incremental, unsupervised algorithm that constructs a topology-preserving representation directly from streaming data: nodes and connecting edges are inserted precisely where the existing representation proves inadequate, and each edge accrues an age value that is reset upon reinforcement and that precipitates pruning should the corresponding connection remain unreinforced beyond a specified threshold.

Adaptive Resonance Theory (Carpenter & Grossberg, 1987) added a similar related idea, the vigilance parameter, governing the degree to which a novel data point is absorbed into an existing category as opposed to creating a wholly new category, thereby forestalling excessive overlap between adjacent regions.

Both algorithms possess an established literature spanning streaming-data clustering, telecommunications network modelling, and pattern recognition; however, neither appears, on the basis of the review conducted in support of this project, to the specific problem of routing across a temporally evolving pool of heterogeneous LLMs.

Scope

In scope:

- A supervised GNG router over 3-5 candidate LLMs
- Labeled dataset construction (RouterBench-style methodology)
- Aging/decay evaluation under simulated churn
- Basic explainability reporting
- Comparative analysis vs 2 other routers

Out of scope:

- Fine-tuning underlying LLMs
- Production deployment infrastructure
- Live long-horizon deployment (churn will be simulated, depending on available time)
- Exhaustive GNG hyperparameter search.

Feasibility

GNG has published pseudocode and open-source implementations; supporting tools (scikit-learn, NumPy) are mature. Resource needs are modest (Local hosting 3-5 LLMs). Ground-truth labels will be generated in-house via the RouterBench methodology.

Dimensionality degradation and possible underperformance versus baselines are the primary project risk— despite which, the project remains reportable and publishable as research.

Technology stack/ algorithm

The system will be primarily designed in Python.

Layer	Tools / Libraries
Embedding generation	OpenAI / Sentence-Transformers embedding models (e.g., text-embedding-3, all-MiniLM, or similar), 384-1536 dimensions
Candidate LLM pool	3-5 heterogeneous specialist and general models mix chosen from HuggingFace
Core routing algorithm	Custom Growing Neural Gas implementation in Python (NumPy/SciPy), extended with edge-aging/decay and label-aware (supervised) growth
Dimensionality handling	scikit-learn (PCA)
Hash / lookup layer	LSH or direct hashing
Baselines for comparison	scikit-learn
Evaluation & benchmarking	RouterBench-style prompt/response dataset, pandas for analysis, matplotlib/seaborn for visualization
Version control & docs	Git/GitHub, Jupyter notebooks for experiments

Algorithm Summary

The processing pipeline proceeds through the following sequence of stages:

1. Each training prompt is embedded by means of a fixed embedding model
2. Every candidate LLM is executed upon every prompt so as to determine the best-performing model per prompt, thereby generating the requisite ground truth. In our implementation, we use specialists to help reduce the need to run each answer, since a general classic choice can be predetermined.
3. A nascent GNG topology is initialized, and nodes are incrementally inserted, employing per-region label impurity, that is, disagreement as to which model prevails, as the growth-triggering error signal, in a manner consonant with the general growing and aging mechanism articulated by Fritzke (1995).
4. An age value is maintained upon each edge, reset upon reinforcement and incremented in its absence, with edges whose age exceeds a specified maximum subject to pruning, in accordance with the standard GNG decay rule.
5. Once the training process has stabilized, each resultant stable region, or leaf, is compiled into a hash table entry mapping the region identifier to the best-performing model together with associated metadata such as mean confidence, task type, and cost.
6. Finally, at the point of inference, the incoming prompt is embedded, the nearest region is identified by means of traversal or direct hashing, and the associated model is returned together with the stored explanatory metadata.

Conclusion

This project applies an established algorithm (GNG) to a specific, underexplored problem: churn-adaptive, explainable routing across independently-trained LLMs.
No prior system combining incremental geometric growth, built-in decay, and explainability for this problem was identified in the literature reviewed.

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